

TUNABLE NON-LINEAR MANIFOLD ROMs FOR ELIPTIC, PARABOLIC, AND HYPERBOLIC PDES VIA MATRIX-FREE PETROV–GALERKIN PROJECTION

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ABSTRACT

A reduced-order PDE solver should let a practitioner choose accuracy and cost after training. We present a nonlinear-manifold reduced-order model (NM-ROM) that provides this choice from one frozen decoder: nested correction directions enlarge the trial space without retraining, while solver tolerance and empirical quadrature set the cost of the reduced solve. The architecture combines a learned spatial bank with a small nonlinear coefficient map, exact boundary enforcement and matrix-free weak residual projection; linear operators are preassembled and validated quadrature reduces nonlinear residual evaluation. Every comparison pairs an accurate and a fast setting of the same model with a named full-order solver in one GPU allocation. On two-dimensional Poisson at 1024^2 the accurate setting reaches 0.96 % error $13.9\times$ faster than conjugate gradients, and the fast setting is $15.2\times$ faster; on an L-shaped domain the accurate setting is $6.07\times$ faster; on held-out three-dimensional Poisson it reaches 0.26 % at $1.33\times$. On Burgers the fast setting is $2.02\times$ faster than Newton–BiCGStab at 1024^2 , and a correction ladder lowers same-grid error $2.44\times$ for $5.16\times$ the runtime; the accurate Burgers setting is slower than the full-order solver up to 1024^2 . Fast-setting speedup grows with resolution in every problem measured. The named solvers remain more accurate at their tolerances. Three-dimensional Burgers, Navier–Stokes and the full wave state miss their targets and are reported as the method’s present limits.

1 INTRODUCTION

Neural operators for partial differential equations (PDEs) (Li et al., 2021; Lu et al., 2021; Kovachki et al., 2023; Li et al., 2024; Boullé and Townsend, 2024) provide fast amortised inference. At a fixed grid and inference procedure, a trained operator returns one accuracy and runtime point. A practitioner may instead need to spend more computation on a difficult query, or accept a less accurate solution when response time matters. Classical full-order methods (FOMs) expose numerical controls such as iteration tolerance, but their cost can grow substantially with mesh resolution. The question is whether a trained reduced model can provide a useful accuracy–cost tradeoff at deployment.

Projection-based reduced-order models address this question by solving the governing equations in a smaller space (Benner et al., 2015; Sirovich, 1987). A linear subspace is inexpensive to use, but can require many basis functions to represent moving fronts. Nonlinear-manifold ROMs learn a more compact representation (Lee and Carlberg, 2020; Kim et al., 2022); their online nonlinear solve can, however, offset the benefit of fewer unknowns. Representation accuracy and computational advantage therefore need to be measured separately.

We present an NM-ROM with a deployment-time accuracy–cost family from one trained decoder. The solution is represented by a learned spatial bank multiplied by a nonlinear map of latent variables. We augment this map with nested, precomputed correction directions. Activating more directions enlarges the trial space without changing the learned weights. The correction rank is the primary representation control; iteration limits and stopping tolerances govern the numerical solve, and empirical quadrature reduces residual-evaluation cost where its validation permits. All online coefficients are determined from the weak PDE residual, without access to the reference solution.

Our experiments measure what this mechanism delivers against full-order solvers. Table 1 reports, for every problem and mesh, the error and the speedup of an accurate and a fast setting of one frozen model against a named solver timed in the same allocation; Figure 1 shows how the speedup changes with resolution, and Table 3 lists the problems on which the method misses its target.

Contributions.

1. **A tunable NM-ROM from one trained model.** Nested correction directions provide discrete deployment settings that change approximation capacity without retraining. We isolate correction rank at fixed residual test count and separately evaluate a prescribed test-count schedule across seeds and held-out cases.
2. **An architecture supporting the accuracy–cost tradeoff.** A learned spatial bank, a nonlinear head with a linear skip, and nested corrections support compact latent solves, boundary enforcement and node-local evaluation. The nonlinear coordinates and linear corrections have distinct roles in the reduced representation.
3. **A practical weak-form implementation and paired evaluation.** Precomputed linear operators, matrix-free differentiation and validated empirical quadrature make the online solve measurable against full-order solvers, and paired accuracy and runtime measurements show where the reduced solve is faster.

2 RELATED WORK

Projection-based and nonlinear-manifold ROMs. Linear ROMs approximate solutions in a fixed subspace (Benner et al., 2015; Sirovich, 1987); moving fronts can require a large basis (Cohen and DeVore, 2015; Greif and Urban, 2019). Nonlinear-manifold ROMs combine learned representations with residual projection (Lee and Carlberg, 2020; Kim et al., 2022), including quadratic manifolds (Geelen et al., 2022; Barnett and Farhat, 2022), neural-field representations (Kim et al., 2024) and adaptive bases (Peherstorfer and Willcox, 2015; Carlberg, 2015). Our focus is a nested set of correction directions within one learned bank, selected after training. Least-squares Petrov–Galerkin projection (Carlberg et al., 2011) and preassembly of reduced operators (Ștefănescu and Sandu, 2014; Weder et al., 2024) are established components; the experiment isolates the effect of correction rank.

Neural operators and hyper-reduction. FNO (Li et al., 2021), DeepONet (Lu et al., 2021), U-Net (Ronneberger et al., 2015; Takamoto et al., 2022) and Transolver (Wu et al., 2024) learn solution maps. Their evaluation resolution or inference procedure can also change accuracy and cost; our mechanism instead changes the trial space used to solve the PDE. Empirical quadrature (Hernández et al., 2017; Yano and Patera, 2019) reduces nonlinear residual evaluation through non-negative weights. We retain that construction, testing its error on states reached by the solver and through independent re-draws. The main iterative linear-system comparator is conjugate gradient (Hestenes and Stiefel, 1952).

3 METHODOLOGY

We turn a learned manifold into a practical PDE solver through three components: a trial manifold whose decoder enforces Dirichlet conditions exactly and carries nested linear correction directions (§3.1); a least-squares Petrov–Galerkin projection of the discrete residual onto fixed weak tests (§3.2); and exact preassembly of every linear term with Empirical Quadrature for Burgers advection where validated (§3.3); §3.4 gives the architecture. Throughout, $u \in \mathbb{R}^n$ is the full-order state on a uniform grid of N intervals per axis, $A \in \mathbb{R}^{n \times n}$ the discrete negative Laplacian; R is the bank width, k the latent dimension, M the number of weak tests and m the number of quadrature nodes. The formulas below illustrate scalar Dirichlet problems; periodic vector fields and time-integrator details are specified per PDE in the appendix. Figure B.1 (Appendix B) shows the data flow; Appendix A gives the per-PDE derivations and exit codes.

3.1 TRIAL MANIFOLD WITH EXACT DIRICHLET ENFORCEMENT

We train a decoder $\mathcal{D} : \mathbb{R}^k \rightarrow \mathbb{R}^n$, $k \ll n$, with latent state z and no encoder in the deployed path, separable into a frozen spatial *bank* $G \in \mathbb{R}^{n \times R}$ times a small neural *head* $h_\theta : \mathbb{R}^k \rightarrow \mathbb{R}^R$,

$k \leq R \ll n$, the bank built once per mesh from a random-Fourier-feature coordinate network g_ϕ (Tancik et al., 2020),

$$G_{x,:} = \mu(x) g_\phi(x)^\top, \quad \mu(x) = 16 x_1(1 - x_1) x_2(1 - x_2). \quad (1)$$

To strictly enforce the homogeneous Dirichlet condition on the boundary Γ , we utilize the smooth vanishing factor μ , zero on Γ , folded into every column of the bank, so that $\partial u / \partial z = 0$ on Γ at every resolution (homogeneous data only).

Nested correction directions. The deployed model augments the head’s output with q fixed directions $C_q \in \mathbb{R}^{R \times q}$ and solves for the latent code and the correction coefficients together,

$$u(z, y) = G(h_\theta(z) + C_q y), \quad z \in \mathbb{R}^k, y \in \mathbb{R}^q, 0 \leq q \leq R. \quad (2)$$

The columns of C_q are chosen offline from training data; the Burgers construction uses principal components of the head’s coefficient errors. Per-PDE constructions are stated in Appendix A. The directions are ordered so that C_q is a prefix of $C_{q'}$ for $q < q'$; the ladder is nested and no rung needs retraining. At $q = 0$ the model is the frozen head; at $q = R$ the head is irrelevant and the representation is the full linear span of the bank. For linear PDEs its coefficients can be found by a reduced linear solve; for nonlinear PDEs the equations can remain nonlinear even in this full-bank representation; in between, q moves the reachable set from the head’s image towards the bank’s span, and the solved dimension from k to $k + q$. Two consequences follow: the bank’s projection error is a floor on the achievable error whatever the head or the solver does, so nonlinearity in h_θ buys a smaller *solved* dimension, not an escape from the span; and all x -dependence factors through G , which makes node sampling and exact preassembly available from one decoder.

3.2 PROJECTING THE PDE ONTO THE MANIFOLD

Each PDE we consider gives a discretised residual $r(u)$ that should vanish at the true solution. We substitute the trial manifold $u(z, y)$ into r and project onto a fixed, latent-independent test space. On the Dirichlet square, $P \in \mathbb{R}^{M \times n}$ collects low-frequency tensor-product sine vectors, eigenvectors of the five-point operator, $PA = \Lambda P$. The reduced problem solved online is

$$(z^*, y^*) = \arg \min_{z, y} \frac{1}{2} \|r_w(z, y)\|_2^2, \quad r_w(z, y) = \Lambda_\star^{-1} P r(u(z, y)) \in \mathbb{R}^M, \quad M > k + q, \quad (3)$$

with a diagonal row scaling $\Lambda_\star$ stated per PDE in Appendix A: a least-squares Petrov–Galerkin condition with an explicit test space (Carlberg et al., 2011; Lee and Carlberg, 2020), not tangent Galerkin, overdetermined rather than square, solved by a method appropriate to each PDE’s structure.

Elliptic (Poisson). The Poisson residual is $r(u) = Au - f$, and projection gives $r_w(z, y) = B_0(h_\theta(z) + C_q y) - b_0$, $B_0 = PG$ (Appendix A.1); the projector onto $\text{range}(B_0 C_q)$ does not depend on z , so y is eliminated exactly (Golub and Pereyra, 1973), the iteration stays k -dimensional at every q , and $q = R$ is a linear solve.

Parabolic (heat). The heat semi-discretisation $du/dt = -\kappa Au$ is advanced with Crank–Nicolson; the reduced step substitutes the manifold into the fully discrete equation *before* projecting and solves $z_{n+1} = \arg \min_z \|B_0 h_\theta(z) - D B_0 h_\theta(z_n)\|_2$ (Appendix A.2), a nonlinear least-squares problem, not a linear system; at $q = R$ the step is the linear recurrence (9). Reflective waves use an explicit latent integration instead (Appendix A.4).

Hyperbolic (Burgers). For $u_t + u(u_x + u_y) = \nu \Delta u$ with a sign-upwind stencil and backward Euler (Appendix A.3) the weak residual is nonlinear in the coefficients, so y is not eliminated in closed form; we damp the (z, y) blocks separately inside one Levenberg–Marquardt step (*block-damped* variable projection), which, in the recorded solver comparison, removes the joint LM configuration’s 6 budget exits (Table D.3).

Solver and exits. Each attempt solves the damped normal system $(H + \lambda \text{diag}(\text{diag } H)) \delta = -g$, $H = J^\top J$, $g = J^\top r_w$, accepting the step only if it strictly decreases the residual: damped Levenberg–Marquardt with a monotone test (Marquardt, 1963). Three exit families are recorded and never conflated: the scale-free stationarity measure $\eta(z, y) = \|J^\top r_w\|_2 / (\|J\|_F \|r_w\|_2) \leq \eta_{\text{tol}}$ (4), a

residual threshold, and the stalls, reported as early-stopped, never as converged. No query uses the solution it predicts: the elliptic solve starts from the cached training code nearest the projected source, the Burgers query fits (z, y) to the supplied initial field (Appendix A.6).

3.3 HYPER-REDUCTION

The non-linear manifold reduces the DOF count from n to $k + q$, but a projected term such as $PN(Gc)$ still costs $O(n)$. For a fixed linear operator the tested residual is exactly precomputable, $P(Au - f) = B(h_\theta(z) + C_q y) - b$ with $B = \Lambda PG$ assembled offline by the corresponding discrete transforms, so linear terms are never approximated; the Burgers advection $PN(Gc)$ is the only term that resists preassembly, evaluated *densely*, $O(nR)$ per residual, or on an *empirical quadrature* rule of m nodes (Hernández et al., 2017; Yano and Patera, 2019), $O(mR)$. The rule is a non-negative *per-node* weight vector, *independent of the snapshot*, fitted by NNLS (Lawson and Hanson, 1974) to reproduce the projected advection term at n_{fit} stored codes with a hard cap of m nodes (Appendix A.5); changing m re-solves the fit, rules are not nested, and deployment selects among stored rules. *A rule is never accepted on its NNLS fit residual*; we score it by the held-out relative error ρ of the projected advection term (14) over states the solver actually reaches on trajectories disjoint from the fit and evaluation cases, against a primary bar $\rho_{\text{max}} \leq 0.116$ fixed before any certification job ran and a tight bar 0.06, and then re-draw the construction: a rule is *confirmed* only if every re-draw passes (§6.3).

3.4 MODEL ARCHITECTURE

Three properties of the problem motivate our architecture: the latent iteration is initialised cold, so the decoder Jacobian must retain a well-conditioned linear component, which motivates a *linear skip* in the head; hyper-reduction retains a sparse subset of mesh nodes, so the decoder must evaluate at any node in mesh-independent time, which motivates a *per-node bank*; and the family must reach from the head’s image to the bank’s whole span without retraining, which motivates the nested *correction directions*.

Linear skip, for cold-start convergence. The head is a two-layer SiLU MLP plus a linear skip, $h_\theta(z) = \varphi_\theta(z) + W^\top z$, so its Jacobian retains a latent-independent component along which the cold solve can descend; the elliptic solver checks the numerical rank of the projected Jacobian at every query. The skip is a design choice, not ablated here.

No encoder; per-node bank. No encoder is needed: the query fits (z, y) against the supplied input. The decoder restricted to a rule’s support is a cached block times the coefficients, and the network’s parameters do not depend on the mesh.

What is fixed, what is chosen, and how error is reported. Every operating point uses one frozen artefact per PDE; selected at run time are the rank q , the quadrature (dense or a stored rule) and the stopping tolerance and budget; M is held fixed along the headline ladder. The representation ablations distinguish three quantities: the *bank floor* (projection error onto range G), the *best-found* error (the smallest any point of the augmented manifold attains, from a multistart oracle) and the *solved* error the iteration returns.

4 IMPLEMENTATION

4.1 MATRIX-FREE PROJECTED OPERATORS VIA JAX

Every linear term is preassembled once per mesh as the $M \times R$ matrix B , so the online residual and Jacobian of a linear PDE are $B(h_\theta(z) + C_q y) - b$ and $B[Dh_\theta \ C_q]$, with Dh_θ from a forward-mode Jacobian-Vector Product (Bradbury et al., 2018); the sampled Burgers advection uses the cached stencil block. The damped normal system is solved directly; no Krylov solve is applied to the projected operator. The Burgers runs above 1024^2 solve it by Cholesky factorisation, clip the step, carry the damping over between time steps and start each step from a quadratic predictor.

4.2 TRAINING PROTOCOL

The bank and the head are trained in two stages, both as auto-decoders: the coordinate network g_ϕ is fitted to the training states through (1), then frozen, and the head is fitted with a code library Z on the bank-projected states; C_q is then constructed from training-only coefficient directions as specified for each PDE. Sizes and cohorts are in Appendix C; checkpoint hashes and recorded offline costs are retained with the source evidence described there.

5 EXPERIMENTAL SETUP AND BENCHMARK PROBLEMS

Problems and references. The two-dimensional problems are Poisson on the square and on an L-shaped domain, heat, viscous Burgers and reflective waves; the three-dimensional problems are Poisson, heat, Burgers and incompressible Navier–Stokes. Table C.1 gives the two-dimensional meshes and cohorts, Table C.2 the sampled families and Table C.5 the three-dimensional configurations. Burgers provides the correction-rank study.

Accuracy. We report worst relative L^2 error over the stated cases and requested times, in percent. Poisson uses the steady reference norm and heat the current reference norm; Burgers and Navier–Stokes use the initial-field norm over evolved times; waves use the energy-state error over displacement and velocity. Same-grid error measures reduction error relative to the converged discrete solution. Error against a refined reference also includes discretisation error; the two are never interchanged. Settings for a held-out (final) cohort are fixed before its cases are accessed; all other cohorts are development cohorts.

Timing and speedup. Every accuracy–runtime pair comes from the same solver invocation. We use double precision, GPU burn-in, synchronised timing and medians of retained repetitions. GPU query time includes initialization, the solve or rollout, and requested device outputs. Complete-query time additionally includes host transfers and is labelled separately. Training and compilation are offline costs. Speedup is $S = T_{\text{FOM}}/T_{\text{method}}$ with both times from the same allocation; each table names the FOM algorithm and shows its error. Solves that miss their stopping rule are reported as such and do not enter a speedup.

Baselines. Linear problems are compared with conjugate gradients (CG), including the L-shaped domain and the CG solve inside each Crank–Nicolson (CN) heat step. Burgers is compared with Newton–BiCGStab and Navier–Stokes with its CNAB2 time integrator. The named FOM of a row is that solver at its fastest tested setting that is at least as accurate as the accurate NM-ROM setting.

6 NUMERICAL EXPERIMENTS AND RESULTS

Table 1 is the headline comparison; the subsections that follow give the settings behind it and the controls that move a frozen model between them.

6.1 ACCURACY AND SPEED AGAINST FULL-ORDER SOLVERS

Linear problems. On Poisson the accurate setting reaches 0.96 % error and is $13.9\times$ faster than CG at 1024^2 ; the corrections cost little speed because they are eliminated analytically (§3.2). The error stays at 0.96 % up to 4096^2 while the GPU-query speedup rises to $146\times$. On the L-shaped domain, where no fast transform applies, the accurate setting is $6.07\times$ faster at 2.12 %. The earlier heat checkpoint crosses CG between 256^2 and 1024^2 ($4.85\times$ at 1024^2). A wider heat bank, evaluated once on a sealed held-out cohort, reaches 0.49 % over all times and is $36.0\times$ faster than CN–CG at 4096^2 ($103\times$ with the batched fit). In three dimensions the held-out Poisson cohort gives 0.26 % at $1.33\times$; three-dimensional heat reaches 0.75 % over evolved times and is slower than CG at both meshes.

Burgers. The reduced solve costs the same at every mesh, so the fast setting passes Newton–BiCGStab at 512^2 and is $2.02\times$ faster at 1024^2 with 2.29 % error. The accurate setting reaches 0.51 % at 256^2 and is slower than the FOM up to 1024^2 (§6.3 gives the cost of each rank). At 4096^2

Table 1: NM-ROM against the named full-order solver at every measured problem and mesh. **Both NM-ROM columns of a row come from one frozen model:** *fast* is the uncorrected setting ($q = 0$) and *accurate* the largest stored correction rank; no network is retrained between them. Error is the worst relative L^2 error over the cohort (%). Speedup is the FOM’s median time divided by the NM-ROM’s in the same GPU allocation, **bold** where the NM-ROM is faster. The FOM is the named iterative solver at its fastest tested setting that is at least as accurate as the accurate setting. ^f held-out final cohort (all others development); ^r error against a refined reference, which includes discretisation error (all others same-grid); ^t all output times including $t = 0$; ^e evolved output times only; ^c complete-query time with host transfers (all others GPU query; heat was measured in GPU query only). Heat (wide bank) is a separately trained frozen model with a 128-function bank ($q = 32$ accurate); over evolved times its accurate error is 0.22 % (0.21 % batched). The batched fit solves each output time independently against exactly propagated test moments, exploiting the linear autonomous structure with eigenfunction tests that the sine-transform solve also uses. With $t = 0$ the Heat 3D model’s accurate error is 1.93 % on its final cohort at 128^3 (Table C.4). The earlier Burgers model has one setting, may exit on a stall, and is shown against relaxed ($10^{-2}/0.5$) and tight ($10^{-6}/10^{-8}$) nonlinear/linear Newton tolerances; the other Burgers rows use the fastest Newton setting at least as accurate (tight-Newton ratios for 2048² and 4096² are in Table C.3). ^h 64 held-out cases never used for selection (not the sealed final cohort). At 2048² the arm chosen on development cases ($M = 544$) was not run on held-out cases, so that held-out row shows the $M = 1088$ rung; at 4096² the $M = 544$ arm reaches 2.93 % on them. Poisson (dev. sources) rows use development sources of a separate run and are not the held-out cohort. At 2048² and 4096² host transfer of the field dominates the NM-ROM query: the accurate setting’s complete-query speedups are $28.7\times$ and $41.1\times$. Dashes mark settings not measured. Times and settings: Table C.3.

Problem	Mesh	NM-ROM accurate		NM-ROM fast		Full-order model (FOM)	
		Err. (%)	Speedup	Err. (%)	Speedup	Err. (%)	Solver
<i>Two-dimensional</i>							
Poisson	256 ²	0.97	3.13 ×	3.16	3.33 ×	0.15	CG, rtol 10 ^{−2}
	1024 ²	0.96	13.9 ×	3.15	15.2 ×	0.072	CG, rtol 10 ^{−2}
	2048 ²	0.96	72.8 ×	3.15	74.3 ×	0.049	CG, rtol 10 ^{−2}
	4096 ²	0.96	146 ×	3.15	148 ×	0.038	CG, rtol 10 ^{−2}
Poisson, L-shape ^c	256 ²	2.13	3.84 ×	3.86	4.04 ×	0.64	CG, rtol 10 ^{−2}
	512 ²	2.12	6.07 ×	3.85	6.18 ×	0.38	CG, rtol 10 ^{−2}
	1024 ²	2.20	9.76 ×	3.85	10.0 ×	0.31	CG, rtol 10 ^{−2}
	2048 ²	2.20	19.3 ×	3.85	19.4 ×	0.24	CG, rtol 10 ^{−2}
Heat ^r	64 ²	—	—	4.56	0.22×	1.59	CN-CG, rtol 10 ^{−2}
	256 ²	—	—	4.56	0.61×	0.93	CN-CG, rtol 10 ^{−2}
	1024 ²	—	—	4.56	4.85 ×	0.77	CN-CG, rtol 10 ^{−2}
Heat (wide bank) ^t	1024 ²	0.24	2.14 ×	0.56	1.81 ×	0.14	CN-CG, Δt=0.05, rtol 10 ^{−3}
Heat (wide bank) ^{t,f}	2048 ²	0.49	9.74 ×	1.36	9.59 ×	0.18	CN-CG, Δt=0.05, rtol 10 ^{−3}
	4096 ²	0.49	36.0 ×	1.36	35.4 ×	0.47	CN-CG, Δt=0.05, rtol 10 ^{−2}
Heat (wide bank, batched fit) ^{t,f}	2048 ²	0.49	60.0 ×	1.36	57.0 ×	0.18	CN-CG, Δt=0.05, rtol 10 ^{−3}
	4096 ²	0.49	103 ×	1.36	101 ×	0.47	CN-CG, Δt=0.05, rtol 10 ^{−2}
Burgers ^e	256 ²	0.51	0.043×	1.89	0.79×	0.049	Newton-BiCGStab, tol 10 ^{−3}
	512 ²	0.55	0.068×	2.14	1.31 ×	0.052	Newton-BiCGStab, tol 10 ^{−3}
	1024 ²	0.59	0.0037×	2.29	2.02 ×	0.034	Newton-BiCGStab, tol 10 ^{−4}
Burgers ^e	2048 ²	0.87	0.99×	2.37	4.16 ×	0.049	Newton-BiCGStab, tol 3×10 ^{−3}
	4096 ²	0.60	4.87 ×	2.41	12.9 ×	0.050	Newton-BiCGStab, tol 3×10 ^{−3}
Burgers (held-out cases) ^{e,h}	2048 ²	1.31	1.43 ×	9.03	5.63 ×	0.13	Newton-BiCGStab, tol 10 ^{−3}
	4096 ²	1.33	5.38 ×	9.03	13.2 ×	0.14	Newton-BiCGStab, tol 3×10 ^{−3}
Burgers (earlier model) ^r	1024 ²	—	—	3.91	1.63 ×	2.39	Newton-BiCGStab, relaxed
	1024 ²	—	—	3.91	14.5 ×	2.14	Newton-BiCGStab, tight
<i>Three-dimensional</i>							
Poisson ^f	32 ³	0.26	0.94×	1.40	0.99×	0.16	CG, rtol 10 ^{−2}
	64 ³	0.26	1.33 ×	1.39	1.38 ×	0.11	CG, rtol 10 ^{−2}
Poisson (dev. sources)	128 ³	0.16	6.75 ×	0.55	6.98 ×	0.075	CG, rtol 10 ^{−2}
	256 ³	0.16	23.2 ×	0.55	23.7 ×	0.049	CG, rtol 10 ^{−2}
Heat ^{e,f}	32 ³	0.76	0.066×	1.56	0.13×	0.32	CN-CG, rtol 10 ^{−4}
	64 ³	0.75	0.13×	1.55	0.27×	0.34	CN-CG, rtol 10 ^{−4}

the accurate setting chosen on six development cases reaches 0.60 % and is $4.87\times$ faster than the fastest Newton setting at least as accurate; on 64 held-out cases the same setting gives 1.33 % at $5.38\times$.¹

¹A restricted-grid recomputation of the held-out errors disagreed with the full-grid value by more than 5 % on 5/384 (2048²) and 18/1152 (4096²) rows; per-arm cohort worsts agree within 1.2 % and the full-grid recomputation is exact.

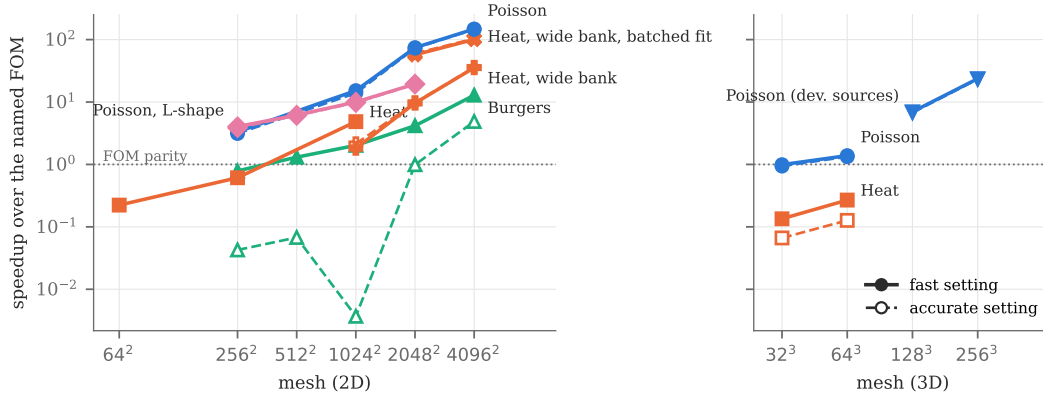


Figure 1: Speedup over the named FOM against mesh, plotted from the rows of Table 1. The accurate Burgers setting at 1024^2 evaluates the residual densely because no quadrature rule is validated there at $q = 256$; with the validated $q = 128$ rule the error is 1.04 % at $0.50\times$. Burgers points at 2048^2 and 4096^2 are the six development cases; held-out errors are higher and are not plotted (Table 1).

Table 2: Comparison with other nonlinear-manifold ROMs on shared data (reserved; rows are filled from the audited comparison run).

Problem, mesh	Method	Err. (%)	Speedup vs. FOM
—	This work (accurate / fast)	—	—
—	Convolutional-autoencoder NM-ROM	—	—
—	Shallow-masked-autoencoder NM-ROM	—	—

Resolution. Figure 1 shows the pattern common to all problems: the NM-ROM query time grows more slowly with the mesh than the iterative FOM’s, so fast-setting speedup rises with resolution in every series.

Where the method currently fails. Table 3 collects the three problems that miss their targets; they are excluded from Table 1. On three-dimensional Burgers the corrections cut the held-out worst error from 16.26 % to 4.40 %, but the FOM is more accurate and faster, and the refined-reference check fails, so only same-grid reduction error is established. On three-dimensional Navier–Stokes the corrections lower the worst error only from 20.34 % to 18.92 %; 31 of 32 held-out cases miss the 5 % target. On reflective waves the fast setting is faster than CG but its energy-state error, which includes velocity, is 11.34 %; corrections bring it to 5.12 %, still above CG’s, at a cost that removes the speedup. Three-dimensional heat misses 1 % once $t = 0$ is counted: its bank represents the initial field only to 1.2–1.9 % at 128^3 – 256^3 . The three-dimensional solves evaluate the residual densely.

6.2 BEST CONFIGURATIONS PER PROBLEM AND RESOLUTION

Table C.3 lists, for every row of Table 1, the correction rank and residual evaluation of both settings with their measured times. Moving between them changes only deployment arguments. Table 4 shows the intermediate settings on Burgers with the residual test space held fixed: each added block of correction directions lowers the error and raises the cost.

6.3 WHICH KNOB TO TURN

Table 5 summarises the deployment-time controls of a frozen model. Correction rank is the accuracy control: with the test space fixed, the Burgers2D ladder lowers the error $2.44\times$ for $5.16\times$ the runtime (with a smaller test space the same ranks give $1.22\times$, so test count is held fixed along the ladder). Empirical quadrature and the stopping tolerance are cost controls: they lower the runtime at little or no change in error (Table 6). The iteration cap is a safeguard, not a control; a truncated solve fails.

Table 3: Problems on which the method misses its target. Error is the worst same-grid relative L^2 error (%) over evolved times; the wave error is the energy-state error over displacement and velocity. Speedup is against the named FOM in the same allocation.

Problem	Setting	Err. (%)	FOM err. (%)	Speedup	FOM
Burgers 3D, 33^3 (final)	$q = 0$	16.26	2.31	$0.038\times$	Newton-BiCGStab
	$q = 192$	4.40	2.31	$0.023\times$	Newton-BiCGStab
Navier-Stokes 3D, 32^3 (final)	$q = 0$	20.34	2.47	$0.0025\times$	CNAB2
	$q = 256$	18.92	2.47	$0.0007\times$	CNAB2
Wave 2D, 1024^2 (dev.)	$q = 0$	11.34	4.29	$3.03\times$	midpoint-CG
	$q = 32$	5.12	4.29	$0.23\times$	midpoint-CG

Table 4: Burgers2D correction settings from one frozen model at 256^2 , test count fixed at $M = 1088$, dense residual. Errors are worst evolved same-grid percentages on development cases; times are paired median GPU query times. All rows meet the stopping rule.

Correction rank q	Relative L^2 error (%)	GPU query time (ms)
0	1.2657	848.0
64	1.0593	1359.4
128	0.8711	1886.5
256	0.5194	4377.9

A separate ladder, in which the test count follows a fixed schedule $M = 4(k + q)$, was evaluated once on a sealed held-out cohort after every choice was frozen. All 4 checkpoints (the original model and three retrained seeds) give monotone error reduction with top-rank errors of 0.59–0.68 %, and 3 of 4 meet the pre-registered secondary criterion, the *knob bar*: converged settings spanning at least $2\times$ in error and $2\times$ in cost. Two stricter pre-registered checks fail: the sealed-to-development error ratio at $q = 0$ for the original model, whose uncorrected solve converges to a wrong branch on one sealed case (10.1120 %, kept in the reported maximum), and universal convergence, which fails on seed2 at $q = 64$ (3 budget exits). Table D.1 retains every checkpoint and verdict.

Limitations. Speedups are against the named iterative solvers, and the named FOM is more accurate than the NM-ROM in every row. On the square and cube a direct sine-transform solve and a coarse-grid solve at the NM-ROM’s own accuracy are both faster than the NM-ROM ($0.67\text{--}0.92\times$ in complete-query time, $0.039\text{--}0.38\times$ in GPU-query time), and on the L-shape the coarse-grid solve is ($0.57\text{--}0.81\times$). On heat at 4096^2 , coarse-grid CN-CG interpolated to the fine mesh (0.29 % in 2.8 ms), the exact sine-transform solve (7.8 ms) and the linear solve in the learned bank, a baseline (0.13 % in 5.1–8.0 ms), are all faster than every NM-ROM setting. The accurate Poisson errors are close to the frozen bank’s projection floor (0.742 % square, 0.143 % cube), which no correction rank can pass; the L-shape accurate setting (2.20 % against a floor of 0.77 %) is limited by the head. On Burgers a 1024^2 Newton solve interpolated to the fine mesh is faster than the accurate setting at similar error (0.42 % in 78 ms at 2048^2 , 0.63 % in 80 ms at 4096^2), and the development accuracy does not carry to held-out cases: the free solve in the same 512-function bank reaches only 1.70 % on a held-out cohort. Except where marked final, cohorts are small development cohorts, and timings are medians without dispersion. The fixed-test-space rank study uses one checkpoint; the multi-seed study changes the test count with the rank, so it does not replicate that intervention. FOM comparisons do not isolate the nonlinear head’s contribution over the linear span of the same bank. Quadrature rules are validated on a finite set of reached states, not certified globally, and none is used in three dimensions. The reduced nonlinear solve can converge to a wrong branch, and lower same-grid error does not remove discretisation error.

Table 5: Which knob to turn, measured on Burgers2D from frozen models. Rank: fixed- M ladder at 256^2 (Table 4). EQ: dense time over EQ time at $q = 0$ in one allocation per mesh. Tolerance and cap: 32 held-out validation cases (Table D.4), stopping tolerance $10^{-8} \rightarrow 10^{-3}$ and iteration cap 2.

Knob	Turn it to	Effect on error	Effect on cost
Correction rank q	$0 \rightarrow 256$	$2.44\times$ lower	$5.16\times$ higher
Empirical quadrature	dense $\rightarrow$ stored rule	Table 6	$4.77\times$ lower at 256^2 , $38.53\times$ at 1024^2
Stopping tolerance	looser	$6.712 \rightarrow 6.701\%$	35.6% lower
Iteration cap	2 iterations	fails (90.3 %)	—

Table 6: Dense and empirical-quadrature (EQ) residual evaluation on Burgers2D at 256^2 . Error is worst evolved relative L^2 (%); time is median GPU milliseconds from one allocation; the last column is dense time over EQ time. Dashes: no paired dense measurement. Rules at $q = 0, 16, 32$ pass every independent reconstruction; higher-rank rules pass some (Table D.2).

Correction rank q	Dense error (%)	Dense ms	EQ error (%)	EQ ms	Dense / EQ
0	1.8890	292.87	1.8891	59.07	$4.96\times$
16	—	—	1.4270	80.61	—
32	—	—	1.2493	97.70	—
64	1.0843	624.93	1.2275	120.63	$5.18\times$
128	0.8930	1190.49	0.8936	246.87	$4.82\times$
256	0.5194	3915.14	0.5389	722.21	$5.42\times$

7 CONCLUSION AND FUTURE WORK

We presented an NM-ROM whose nested corrections change approximation capacity after training. From one frozen model, the accurate setting is $13.9\times$ faster than CG on Poisson at 1024^2 with 0.96 % error, the fast Burgers setting is $2.02\times$ faster than Newton-BiCGStab at 1024^2 , and fast-setting speedup grows with resolution in every problem measured. Accurate nonlinear solves remain slower than the FOM up to 1024^2 , and three problems miss their targets (Table 3). Future work is a cheaper high-rank nonlinear solve, quadrature in three dimensions, reliable nonlinear initialization and unstructured meshes.

REPRODUCIBILITY STATEMENT

Tables and numerical prose are generated from retained run records by the accompanying scripts. Source hashes, checkpoints, solver settings, cohorts, timing repetitions and audit results identify each comparison (Appendix C). The source archive retains failed settings; the paper distinguishes development selection from held-out evaluation.

AI USE STATEMENT

Language-model assistants supported method and experiment development, implementation, result analysis, auditing, and manuscript drafting and editing. Numerical results were computed by the recorded solver runs; table generators reproduce the reported values from those records. The authors are responsible for verifying the methods, results and text.

REFERENCES

- J. Barnett and C. Farhat. Quadratic approximation manifold for mitigating the Kolmogorov barrier in nonlinear projection-based model order reduction. *Journal of Computational Physics*, 464:111348, 2022.
- P. Benner, S. Gugercin, and K. Willcox. A survey of projection-based model reduction methods for parametric dynamical systems. *SIAM Review*, 57(4):483–531, 2015.

- G. Berkooz, P. Holmes, and J. L. Lumley. The proper orthogonal decomposition in the analysis of turbulent flows. *Annual Review of Fluid Mechanics*, 25(1):539–575, 1993.
- S. Berrone, C. Canuto, M. Pintore, and N. Sukumar. Enforcing Dirichlet boundary conditions in physics-informed neural networks and variational physics-informed neural networks. *arXiv preprint arXiv:2210.14795*, 2022.
- N. Boullé and A. Townsend. A mathematical guide to operator learning. In *Handbook of Numerical Analysis*, volume 25, pages 83–125. Elsevier, 2024.
- J. Bradbury, R. Frostig, P. Hawkins, M. J. Johnson, C. Leary, D. Maclaurin, G. Necula, A. Paszke, J. VanderPlas, S. Wanderman-Milne, and Q. Zhang. JAX: composable transformations of Python+NumPy programs, 2018.
- R. Brecht, R. O. Popovych, A. Bihlo, and Y. Popovych. Improving physics-informed DeepONets with hard constraints. *arXiv preprint arXiv:2309.07899*, 2023.
- H. Cai, C. Gan, T. Wang, Z. Zhang, and S. Han. Once-for-All: Train one network and specialize it for efficient deployment. In *International Conference on Learning Representations*, 2020.
- S. Cao. Choose a Transformer: Fourier or Galerkin. In *Advances in Neural Information Processing Systems*, 2021.
- K. Carlberg, C. Bou-Mosleh, and C. Farhat. Efficient non-linear model reduction via a least-squares Petrov–Galerkin projection and compressive tensor approximations. *International Journal for Numerical Methods in Engineering*, 86(2):155–181, 2011.
- S. Chaturantabut and D. C. Sorensen. Nonlinear model reduction via discrete empirical interpolation. *SIAM Journal on Scientific Computing*, 32(5):2737–2764, 2010.
- W. Chen, J. Song, P. Ren, S. Subramanian, D. Morozov, and M. W. Mahoney. Data-efficient operator learning via unsupervised pretraining and in-context learning. In *Advances in Neural Information Processing Systems*, 2024.
- A. Cohen and R. DeVore. Approximation of high-dimensional parametric PDEs. *Acta Numerica*, 24:1–159, 2015.
- A. N. Diaz, Y. Choi, and M. Heinkenschloss. A fast and accurate domain-decomposition nonlinear manifold reduced order model. *Computer Methods in Applied Mechanics and Engineering*, 425:116943, 2024.
- M. S. Eshaghi, C. Anitescu, N. Valizadeh, Y. Wang, X. Zhuang, and T. Rabczuk. Nows: Neural operator warm starts for accelerating iterative solvers. *arXiv preprint arXiv:2511.02481*, 2025.
- C. Farhat, P. Avery, T. Chapman, and J. Cortial. Dimensional reduction of nonlinear finite element dynamic models with finite rotations and energy-based mesh sampling and weighting for computational efficiency. *International Journal for Numerical Methods in Engineering*, 98(9):625–662, 2014.
- S. Fresca and A. Manzoni. POD-DL-ROM: Enhancing deep learning-based reduced order models for nonlinear parametrized PDEs by proper orthogonal decomposition. *Computer Methods in Applied Mechanics and Engineering*, 388:114181, 2022.
- R. Geelen, S. Wright, and K. Willcox. Operator inference for non-intrusive model reduction with quadratic manifolds. *Computer Methods in Applied Mechanics and Engineering*, 403:115717, 2022.
- C. Greif and K. Urban. Decay of the Kolmogorov N -width for wave problems. *Applied Mathematics Letters*, 96:216–222, 2019.
- A. Griewank and A. Walther. *Evaluating Derivatives: Principles and Techniques of Algorithmic Differentiation*. SIAM, Philadelphia, 2nd edition, 2008.

- Z. Hao, Z. Wang, H. Su, C. Ying, Y. Dong, S. Liu, Z. Cheng, J. Song, and J. Zhu. GNOT: A general neural operator transformer for operator learning. In *International Conference on Machine Learning*, 2023.
- Z. Hao, C. Su, S. Liu, J. Berner, C. Ying, H. Su, A. Anandkumar, J. Song, and J. Zhu. DPOT: Auto-regressive denoising operator transformer for large-scale PDE pre-training. In *International Conference on Machine Learning*, 2024.
- M. Herde, B. Raonic, T. Rohner, R. Käppeli, R. Molinaro, E. de Bezenac, and S. Mishra. Poseidon: Efficient foundation models for PDEs. In *Advances in Neural Information Processing Systems*, 2024.
- J. A. Hernández, M. A. Caicedo, and A. Ferrer. Dimensional hyper-reduction of nonlinear finite element models via empirical cubature. *Computer Methods in Applied Mechanics and Engineering*, 313:687–722, 2017.
- M. R. Hestenes and E. Stiefel. Methods of conjugate gradients for solving linear systems. *Journal of Research of the National Bureau of Standards*, 49(6):409–436, 1952.
- J. S. Hesthaven, G. Rozza, and B. Stamm. *Certified Reduced Basis Methods for Parametrized Partial Differential Equations*. SpringerBriefs in Mathematics. Springer, 2016.
- M. Hirsch, F. Pichi, and J. S. Hesthaven. Neural empirical interpolation method for nonlinear model reduction. *arXiv preprint arXiv:2406.03562*, 2024.
- B. Holzschuh, Q. Liu, G. Kohl, and N. Thuerey. PDE-Transformer: Efficient and versatile transformers for physics simulations. In *International Conference on Machine Learning*, 2025.
- W. Hu, S. Liu, P. Qiao, Z. Sun, and Y. Dou. Transolver is a linear transformer: Revisiting physics-attention through the lens of linear attention. *arXiv preprint arXiv:2511.06294*, 2025.
- T. J. R. Hughes. *The Finite Element Method: Linear Static and Dynamic Finite Element Analysis*. Dover Publications, 2000.
- G. E. Karniadakis, I. G. Kevrekidis, L. Lu, P. Perdikaris, S. Wang, and L. Yang. Physics-informed machine learning. *Nature Reviews Physics*, 3:422–440, 2021.
- Y. Kim, Y. Choi, D. Widemann, and T. Zohdi. A fast and accurate physics-informed neural network reduced order model with shallow masked autoencoder. *Journal of Computational Physics*, 451:110841, 2022.
- M. Kim, T. Wen, K. Lee, and Y. Choi. Physics-informed reduced order model with conditional neural fields. *arXiv preprint arXiv:2412.05233*, 2024.
- D. Kochkov, J. A. Smith, A. Alieva, Q. Wang, M. P. Brenner, and S. Hoyer. Machine learning-accelerated computational fluid dynamics. *Proceedings of the National Academy of Sciences*, 118(21):e2101784118, 2021.
- A. Kopaničáková and G. E. Karniadakis. DeepOnet based preconditioning strategies for solving parametric linear systems of equations. *arXiv preprint arXiv:2401.02016*, 2024.
- N. Kovachki, Z. Li, B. Liu, K. Azizzadenesheli, K. Bhattacharya, A. Stuart, and A. Anandkumar. Neural operator: learning maps between function spaces. *Journal of Machine Learning Research*, 24(89):1–97, 2023.
- C. L. Lawson and R. J. Hanson. *Solving Least Squares Problems*. Prentice-Hall, Englewood Cliffs, NJ, 1974.
- K. Lee and K. T. Carlberg. Model reduction of dynamical systems on nonlinear manifolds using deep convolutional autoencoders. *Journal of Computational Physics*, 404:108973, 2020.
- Z. Li, N. Kovachki, K. Azizzadenesheli, B. Liu, K. Bhattacharya, A. Stuart, and A. Anandkumar. Fourier neural operator for parametric partial differential equations. In *International Conference on Learning Representations*, 2021.

- Z. Li, D. Z. Huang, B. Liu, and A. Anandkumar. Fourier neural operator with learned deformations for PDEs on general geometries. *Journal of Machine Learning Research*, 24(388):1–26, 2023.
- Z. Li, K. Meidani, and A. B. Farimani. Transformer for partial differential equations’ operator learning. *Transactions on Machine Learning Research*, 2023.
- Z. Li, H. Zheng, N. Kovachki, D. Jin, H. Chen, B. Liu, K. Azizzadenesheli, and A. Anandkumar. Physics-informed neural operator for learning partial differential equations. *ACM/IMS Journal of Data Science*, 1(3):1–27, 2024.
- P. Lippe, B. Veeling, P. Perdikaris, R. E. Turner, and J. Brandstetter. PDE-Refiner: Achieving accurate long rollouts with neural PDE solvers. In *Advances in Neural Information Processing Systems*, 2023.
- L. Lu, P. Jin, G. Pang, Z. Zhang, and G. E. Karniadakis. Learning nonlinear operators via DeepONet based on the universal approximation theorem of operators. *Nature Machine Intelligence*, 3(3):218–229, 2021.
- M. McCabe, B. Régald-Saint Blancard, L. H. Parker, R. Ohana, M. Cranmer, A. Bietti, M. Eickenberg, S. Golkar, G. Krawezik, F. Lanasse, M. Pettee, T. Tesileanu, K. Cho, and S. Ho. Multiple physics pretraining for physical surrogate models. In *Advances in Neural Information Processing Systems*, 2024.
- M. A. Mirhoseini and M. J. Zahr. Accelerated solutions of convection-dominated partial differential equations using implicit feature tracking and empirical quadrature. *arXiv preprint arXiv:2305.15661*, 2023.
- F. Negri, A. Manzoni, and D. Amsallem. Efficient model reduction of parametrized systems by matrix discrete empirical interpolation. *Journal of Computational Physics*, 303:431–454, 2015.
- N. C. Nguyen. High-order empirical interpolation methods for real time solution of parametrized nonlinear PDEs. *arXiv preprint arXiv:2410.02100*, 2024.
- J. Nocedal and S. J. Wright. *Numerical Optimization*. Springer Series in Operations Research and Financial Engineering. Springer, New York, 2nd edition, 2006.
- M. Ohlberger and S. Rave. Reduced basis methods: success, limitations and future challenges. In *Proceedings of the ALGORITHM Conference*, 2016.
- A. Quarteroni, A. Manzoni, and F. Negri. *Reduced Basis Methods for Partial Differential Equations: An Introduction*. UNITEXT, Vol. 92. Springer, 2015.
- M. Raissi, P. Perdikaris, and G. E. Karniadakis. Physics-informed neural networks. *Journal of Computational Physics*, 378:686–707, 2019.
- F. Romor, G. Stabile, and G. Rozza. Non-linear manifold reduced-order models with convolutional autoencoders and reduced over-collocation method. *Journal of Scientific Computing*, 94:74, 2023.
- Y. Saad. *Iterative Methods for Sparse Linear Systems*. SIAM, Philadelphia, 2nd edition, 2003.
- L. J. Shikhman. One operator to rule them all? On boundary-indexed operator families in neural PDE solvers. *arXiv preprint arXiv:2603.01406*, 2026.
- L. Sirovich. Turbulence and the dynamics of coherent structures, Parts I–III. *Quarterly of Applied Mathematics*, 45(3):561–590, 1987.
- N. Sukumar and A. Srivastava. Exact imposition of boundary conditions with distance functions in physics-informed deep neural networks. *Computer Methods in Applied Mechanics and Engineering*, 389:114333, 2022.
- A. Tran, A. Mathews, L. Xie, and C. S. Ong. Factorized Fourier neural operators. In *International Conference on Learning Representations*, 2023.

- K. Um, R. Brand, Y. Fei, P. Holl, and N. Thuerey. Solver-in-the-loop: Learning from differentiable physics to interact with iterative PDE-solvers. In *Advances in Neural Information Processing Systems*, 2020.
- S. Wen, A. Kumbhat, L. Lingsch, S. Mousavi, Y. Zhao, P. Chandrashekar, and S. Mishra. Geometry aware operator transformer as an efficient and accurate neural surrogate for PDEs on arbitrary domains. In *Advances in Neural Information Processing Systems*, 2025.
- T. Wang and C. Wang. Latent neural operator for solving forward and inverse PDE problems. In *Advances in Neural Information Processing Systems*, 2024.
- H. Wang, J. Li, A. Dwivedi, K. Hara, and T. Wu. BENO: Boundary-embedded neural operators for elliptic PDEs. In *International Conference on Learning Representations*, 2024.
- H. Wu, H. Luo, H. Wang, J. Wang, and M. Long. Transolver: A fast transformer solver for PDEs on general geometries. In *International Conference on Machine Learning*, 2024.
- M. Yano and A. T. Patera. An LP empirical quadrature procedure for reduced basis treatment of parametrized nonlinear PDEs. *Computer Methods in Applied Mechanics and Engineering*, 344:1104–1123, 2019.
- J. Yu, L. Yang, N. Xu, J. Yang, and T. Huang. Slimmable neural networks. In *International Conference on Learning Representations*, 2019.
- J. Barnett, C. Farhat, and Y. Maday. Neural-network-augmented projection-based model order reduction for mitigating the Kolmogorov barrier to reducibility. *Journal of Computational Physics*, 492:112420, 2023.
- K. Carlberg, C. Farhat, J. Cortial, and D. Amsallem. The GNAT method for nonlinear model reduction: Effective implementation and application to computational fluid dynamics and turbulent flows. *Journal of Computational Physics*, 242:623–647, 2013.
- K. Carlberg. Adaptive h -refinement for reduced-order models. *International Journal for Numerical Methods in Engineering*, 102(5):1192–1210, 2015.
- T. Chapman, P. Avery, P. Collins, and C. Farhat. Accelerated mesh sampling for the hyper reduction of nonlinear computational models. *International Journal for Numerical Methods in Engineering*, 109(12):1623–1654, 2017.
- G. H. Golub and V. Pereyra. The differentiation of pseudo-inverses and nonlinear least squares problems whose variables separate. *SIAM Journal on Numerical Analysis*, 10(2):413–432, 1973.
- D. W. Marquardt. An algorithm for least-squares estimation of nonlinear parameters. *Journal of the Society for Industrial and Applied Mathematics*, 11(2):431–441, 1963.
- B. Peherstorfer and K. Willcox. Online adaptive model reduction for nonlinear systems via low-rank updates. *SIAM Journal on Scientific Computing*, 37(4):A2123–A2150, 2015.
- O. Ronneberger, P. Fischer, and T. Brox. U-Net: Convolutional networks for biomedical image segmentation. In *Medical Image Computing and Computer-Assisted Intervention (MICCAI)*, pages 234–241, 2015.
- R. Ștefănescu and A. Sandu. Efficient approximation of sparse Jacobians for time-implicit reduced order models. *arXiv preprint arXiv:1402.2018*, 2014.
- P. N. Swarztrauber. The methods of cyclic reduction, Fourier analysis and the FACR algorithm for the discrete solution of Poisson’s equation on a rectangle. *SIAM Review*, 19(3):490–501, 1977.
- M. Takamoto, T. Praditia, R. Leiteritz, D. MacKinlay, F. Alesiani, D. Pflüger, and M. Niepert. PDEBench: An extensive benchmark for scientific machine learning. In *Advances in Neural Information Processing Systems (Datasets and Benchmarks Track)*, volume 35, 2022.
- M. Tancik, P. P. Srinivasan, B. Mildenhall, S. Fridovich-Keil, N. Raghavan, U. Singhal, R. Ramamoorthi, J. T. Barron, and R. Ng. Fourier features let networks learn high frequency functions in low dimensional domains. *Advances in Neural Information Processing Systems*, 33, 2020.

J. Weder, P. Schwerdtner, and B. Peherstorfer. Online-efficient model reduction on quadratic manifolds with Neural Galerkin schemes. *arXiv preprint arXiv:2412.17695*, 2024.

A METHOD DETAILS

The generic overdetermined problem (3) is instantiated differently per PDE. Poisson, heat and Burgers below are fully discrete weak least-squares problems; the wave arm (§A.4) is an explicit second-order latent integration and is not an instance of (3). Formulas are for the scalar two-dimensional Dirichlet case; three-dimensional and periodic-vector scope is stated in Appendix C.1.

A.1 ELLIPTIC INSTANCE: POISSON

For $Au = f$ we take $\Lambda_\star = \Lambda$, so the weak residual is

$$r_w(z, y) = \Lambda^{-1}P(Au - f) = B_0(h_\theta(z) + C_q y) - b_0, \quad B_0 = PG, \quad b_0 = \Lambda^{-1}Pf. \quad (5)$$

Because the retained tests are orthonormal and $Au^\star = f$, this residual is exactly the tested error $P(u - u^\star)$, so the minimised objective is the squared discrete L^2 error of the reduced state in the retained modes; this holds only for the modes P retains.

Analytically eliminated corrections. Let $B_0 C_q = QR$ be a thin QR factorisation. Q does not depend on z , so the inner minimisation over y has the closed form $\mathcal{R}y = Q^\top(b_0 - B_0 h_\theta(z))$ and the minimised value is

$$\min_y \|B_0(h_\theta(z) + C_q y) - b_0\| = \|B_\perp h_\theta(z) - b_\perp\|, \quad B_\perp = (I - QQ^\top)B_0, \quad b_\perp = (I - QQ^\top)b_0. \quad (6)$$

The LM iteration runs on z alone with the deflated operator, and y is recovered by one triangular solve. Because the projector is constant the elimination introduces no approximation and $B_\perp D h_\theta$ is the exact Jacobian of the eliminated residual; the only dropped derivative anywhere in the elliptic solve is the head curvature. At $q = R$ the deflated operator is zero, the latent problem is degenerate by construction, and the model is the linear least-squares solve $\min_y \|B_0 y - b_0\|$; the timed top rung is that QR solve, not an LM iteration on a zero Jacobian. We report the reduced stationarity of the deflated problem, the augmented stationarity of the full problem, the backward error of the triangular recovery and the numerical rank of $B_0 C_q$.

A.2 PARABOLIC INSTANCE: HEAT

The semi-discrete heat equation $\dot{u} = -\kappa Au$ is advanced with Crank–Nicolson, $(I + \frac{\Delta t \kappa}{2} A)u^{n+1} = (I - \frac{\Delta t \kappa}{2} A)u^n$, and the reduced step substitutes the manifold into the fully discrete equation *before* projecting. With $u^n = u(z_n)$, projecting with P and row-scaling by $\Lambda_\star = I + \frac{\Delta t \kappa}{2} \Lambda$ gives

$$r_{w,n}(z) = B_0 h_\theta(z) - D B_0 h_\theta(z_n), \quad D = \text{diag}\left(\frac{1 - \frac{\Delta t \kappa}{2} \lambda_i}{1 + \frac{\Delta t \kappa}{2} \lambda_i}\right), \quad (7)$$

and $z_{n+1} = \arg \min_z \|r_{w,n}(z)\|_2$, warm-started at z_n . Two points are structural. The right-hand side is built from the *decoded current state* at every step: carrying the quantity the previous step already made small freezes the recursion and reproduces a one-step solution to round-off. And z_{n+1} enters through h_θ , so the step is a nonlinear least-squares problem, not a linear solve. With corrections, $c_n = h_\theta(z_n) + C_q y_n$ replaces $h_\theta(z_n)$:

$$(z_{n+1}, y_{n+1}) = \arg \min_{z, y} \|B_0(h_\theta(z) + C_q y) - D B_0 c_n\|_2, \quad (8)$$

where y is eliminated as in (6) and the *complete* previous coefficients c_n , corrections included, are carried into the next step. At $q = R$ the head is redundant and the step is the linear least-squares recurrence

$$c_{n+1} = B_0^+ D B_0 c_n, \quad (9)$$

with B_0^+ the pseudo-inverse ($M \geq R$; the numerical rank of B_0 is checked) and c_0 fitted to the supplied initial field. This endpoint is the weak Crank–Nicolson propagation of the bank coefficients, with no head and no iteration. It is the “top rung” of the heat ladder and the linear-bank baseline of the text. The deployed path factors the damped normal matrix by Cholesky; a variant assembles the Gram matrix $S = B_0^\top B_0$ offline and forms $H = D h_\theta^\top S D h_\theta$ directly, which is exact and is reported as an algebraic ablation of the same step.

A.3 NONLINEAR ADVECTION: BURGERS

The full-order problem is $u_t + u(u_x + u_y) = \nu \Delta u$ with the non-conservative sign-upwind stencil

$$N(u)_{ij} = u_{ij} (\delta_x u + \delta_y u)_{ij}, \quad (\delta_x u)_{ij} = N \cdot \begin{cases} u_{ij} - u_{i-1,j}, & u_{ij} > 0, \\ u_{i+1,j} - u_{ij}, & u_{ij} \leq 0, \end{cases} \quad (10)$$

δ_y analogous and switching on the same centre value, ghost zeros on all four walls, and backward Euler in time, $r_n(u) = u - u^n + \Delta t (N(u) - \nu \Delta_h u)$. Projecting and row scaling by $\Lambda_\star = I + \Delta t \nu \Lambda$ gives, with $c = h_\theta(z) + C_q y$,

$$r_{w,n} = (I + \Delta t \nu \Lambda)^{-1} [B_0 c - B_0 c_n + \Delta t (\hat{N}(c) + \nu \Lambda B_0 c)], \quad (11)$$

where the diffusion term is exact and the row scaling is the diagonal of the linearised implicit operator in the modal basis, the modal form of the Helmholtz preconditioner the full-order Newton solver uses. Only $\hat{N}(c) \approx PN(Gc)$ requires a choice: dense evaluation at every interior node, the sampled rule $\hat{N}(c) = \sum_{i \in \mathcal{S}} w_i P_{:,i} N_i(Gc)$ with the sign switch retained at the retained nodes, or the precomputed quadratic form $\frac{1}{2} c^\top Q c$ with Q the symmetrised tensor $T_{iab} = \sum_x P_{ix} G_{xa} (DG)_{xb}$, which is an identity only where $u > 0$ at every stencil node of every trial state the solver visits. The last is an algebraic route we verified in-job (residual and Jacobian parity below 10^{-11} relative on sign-satisfying states) and do not time in this paper. Time stepping uses a fixed substep count per output interval; each step is warm-started at the previous code, and the linear extrapolation is used instead when, and only when, its residual norm is smaller.

Block-damped step. With $J = [J_z \ J_y]$ the Jacobian of (11) in (z, y) , obtained in one forward-mode pass, each iteration solves

$$(J^\top J + \lambda D_z + \varepsilon D_y) \begin{bmatrix} \delta z \\ \delta y \end{bmatrix} = -J^\top r_{w,n}, \quad \|\delta z\| \leq \Delta, \quad (12)$$

where $D_z = \text{diag}(\text{diag}(J_z^\top J_z), 0)$ carries the Levenberg–Marquardt damping on the latent block only, $D_y = \text{diag}(0, I)$ with a fixed small ridge ε , and Δ is the $q = 0$ trust radius. The correction step is therefore an undamped Gauss–Newton step on the current linearisation, not throttled by a radius calibrated for the latent code. The step is accepted only if the residual decreases; otherwise λ grows as in §A.6. Table D.3 compares it with joint damping and with variable projection.

A.4 SECOND-ORDER INSTANCE: REFLECTIVE WAVES

For $u_{tt} = c^2 \Delta u$ the bank is mass-orthonormal and the R weak tests are the bank itself, so projection is Galerkin with stiffness $K = G^\top M_h A G$, M_h the diagonal mass matrix. Writing the coefficients as $a(z, y) = h_\theta(z) + C_q y$ and differentiating twice in time, $\ddot{a} = Dh_\theta \ddot{z} + h''_\theta(z)[\dot{z}, \dot{z}] + C_q \ddot{y}$, the projected equation $\ddot{a} = -c^2 K a$ becomes a linear least-squares problem for the joint acceleration,

$$[Dh_\theta(z) \ C_q] \begin{pmatrix} \ddot{z} \\ \ddot{y} \end{pmatrix} = - (c^2 K a + h''_\theta(z)[\dot{z}, \dot{z}]), \quad (13)$$

whose curvature term $h''_\theta[\dot{z}, \dot{z}]$ is a forward-over-forward derivative of the head. The first-order system in $(z, y, \dot{z}, \dot{y})$ is advanced with classical RK4 at a fixed step, one solve of (13) per stage; there is no implicit residual minimisation and no stationarity exit. Initial coordinates and velocities are obtained from the supplied displacement and velocity. At $q = R$ the head is dropped and $\ddot{a} = -c^2 K a$ is propagated exactly through the eigendecomposition $K = V \Lambda V^\top$, $a(t) = V(\cos(\omega t) \hat{a}_0 + \omega^{-1} \sin(\omega t) \hat{b}_0)$, $\omega = c\sqrt{\Lambda}$. The iterative full-order comparator uses implicit midpoint with CG. The error reported in Table 3 is the energy-state error over displacement and velocity.

A.5 EMPIRICAL QUADRATURE: THE FIT AND THE COUNTS

A fitted rule with support $\mathcal{S}$ and weights w is scored by the held-out relative error of the projected advection term,

$$\rho(u) = \frac{\|\sum_{i \in \mathcal{S}} w_i P_{:,i} N_i(u) - PN(u)\|_2}{\|PN(u)\|_2}, \quad \rho_{\max} \leq 0.116 \text{ (primary bar)}, \quad \rho_{\max} \leq 0.06 \text{ (tight)}, \quad (14)$$

over the held-out reachable states described below; the primary bar is the held-out ρ of the incumbent $q=0$ rule at the state carrying the first-interval penalty, measured in the q-diag cell and adopted as the primary bar in the q-ridge design before any certification job ran; the tight bar was declared before b-eqtop ran. A candidate node set $\mathcal{C}$ is drawn once with a fixed seed. For each of n_{fit} stored codes $c^{(\ell)}$ we evaluate the decoded state, form the per-node contributions $N_i(Gc^{(\ell)})$ and the exact full-grid target $PN(Gc^{(\ell)})$, and solve

$$\min_{w \geq 0} \|\mathcal{G} w - \beta\|_2, \quad \mathcal{G}_{(\ell,i),c} = P_{i,c} N_c(Gc^{(\ell)}), \quad \beta_{(\ell,i)} = [PN(Gc^{(\ell)})]_i, \quad (15)$$

rows normalised by their Euclidean norms, support grown greedily by the Lawson–Hanson criterion with an exact non-negative refit after each addition and a hard cap of m nodes, padded to the requested count if the support saturates. Three counts are separated: the requested node count m , the achieved support, and the number of fit states n_{fit} . In the certification cell the fit states are reachable states (states of the dense solver’s own converged per-step trajectory on fit trajectories), the held-out states are 512 reachable states per rung from certification trajectories disjoint from both the fit and the evaluation cases, and ρ of (14) is evaluated on those. A deployed rule is a stored triple: node indices, weights, and the cached $m \times 5 \times R$ bank block.

A.6 SOLVER CONSTANTS AND EXIT CODES

Damping starts at $\lambda_0 = 10^{-6}$ (Poisson, Burgers) or 10^{-4} (heat); $\lambda \leftarrow \max(\lambda/3, 10^{-12})$ on acceptance and $\min(10\lambda, \lambda_{\text{max}})$ on rejection; the trust radius is taken from the spread of the training codes. Exit codes differ per cell and are reported per cell. Burgers: 4 stationarity, 1 residual, 2 tiny step, 3 rejected at $\lambda \geq 10^{14}$, 0 budget. Poisson: 6 stationarity, 2 relative residual, 1 stall, 3 damping limit, 5 non-finite initial value, 0 budget. Heat: 1 stationarity, 2 tiny step, 3 damping limit, 4 non-finite, 0 budget. The heat criterion divides by $\|J\|_F$ only, applied to a residual already normalised by the target norm; Poisson and Burgers use (4) directly. The completion rule that accepts an attained initial fit (§3.2) was pre-registered in the panel cell’s design before the job ran; complete exit flags remain in the source records.

B ARCHITECTURE AND ONLINE DATA FLOW

C EXPERIMENTAL CONFIGURATION AND REPRODUCIBILITY

Tables C.1 and C.2 identify the two-dimensional studies and Table C.5 the three-dimensional ones. Within each study, training precedes evaluation and the selected bank, head and correction directions remain frozen. Deployment changes reduced coordinates, not network weights. Recorded training configurations and available offline costs are retained with the source evidence; unrecorded training costs are not inferred.

Table C.1: Problem specification. Cohort and reduced sizes are read from the run configurations where recorded. The Burgers sealed cohort has been opened and is reported in Table D.1; other rows describe their recorded development and validation cohorts.

PDE	equation, domain, boundary	meshes	time stepping	reduced sizes	reference	cohorts
Burgers 2D	$u_t + u(u_x + u_y) = \nu \Delta u$, $(0, 1)^2$, $u _{\partial\Omega} = 0$	256^2 (ladder 64–1024)	$\Delta t = 0.005$, backward Euler, sign- upwind	$K = 16$, $R = 512$	refined 4096^2 , $\Delta t = 0.00015625$	6 development cases; 32 held-out (tuning); sealed cohort opened once (job 3804465)
Poisson 2D	$-\Delta u = f$, $(0, 1)^2$, $u _{\partial\Omega} = 0$	256^2 , 1024^2	none (elliptic)	$K = 16$, $R = 128$ (incumbent); $K = 32$, $R = 512$	exact discrete (DST); 2048^2 refinement	12 development sources
Heat 2D	$u_t = \kappa \Delta u$, $(0, 1)^2$	64^2 – 1024^2	Crank–Nicolson	$k = 8$, $R = 32$	refined-grid reference ($1024^2/2048^2$ pair); error includes discreti- sation	12 development cases, 3 repetitions; measured in job 3529772; checkpoint lineage: earlier cell, job 3511417
Wave 2D (reflec- tive)	$u_{tt} = c^2 \Delta u$, $(0, 1)^2$, $u _{\partial\Omega} = 0$	64^2 , 256^2 , 1024^2	RK4 on the manifold; exact modal propaga- tion for the bank none	$K = 32$, $R = 64$	direct DST	8 development cases
Poisson, L-shape	$-\Delta u = f$, $(0, 1)^2 \setminus [\frac{1}{2}, 1)^2$	256^2 , 512^2		$K \in \{16, 32\}$, $R \in \{256, 512, 514\}$	sparse direct (Su- perLU)	3072 / 256 / 32 sources (train / selection / develop- ment)

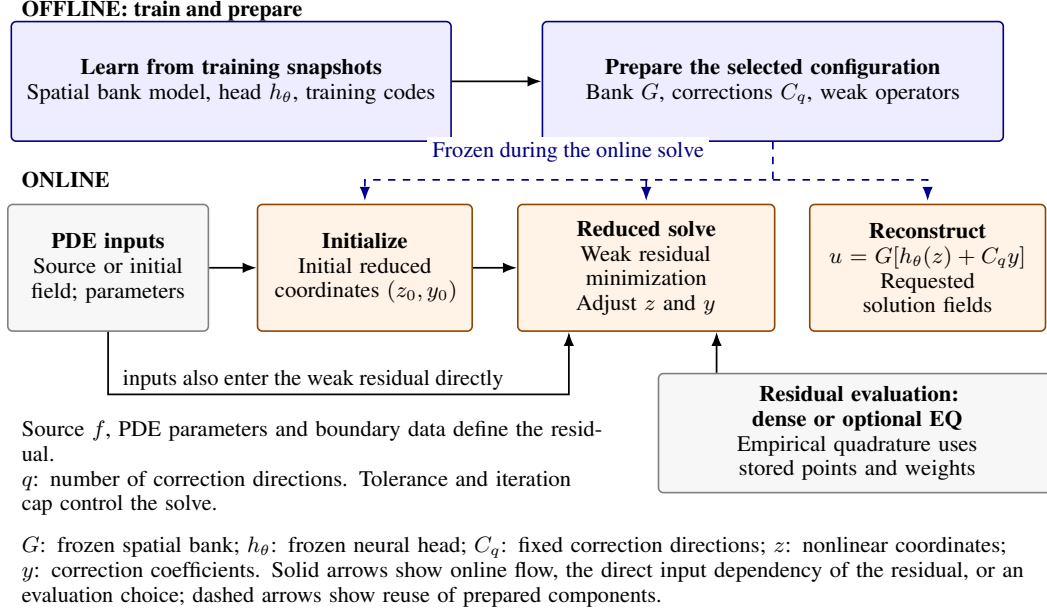


Figure B.1: NM-ROM from training to prediction. Blue components are prepared before the query and remain frozen; orange boxes compute the online solution. The inputs initialize reduced coordinates, which are adjusted to minimize the weak PDE residual before reconstructing the requested fields. Initialization and the reduced solver are PDE-specific; linear-PDE corrections can be eliminated analytically. Time-dependent problems repeat the reduced step, with reconstruction at requested output times. Empirical quadrature (EQ) is an optional residual evaluation, used in the Burgers2D EQ panels; the current 3D and wave panels do not use it. Correction rank changes the representation, whereas EQ changes residual evaluation. Baselines are evaluated independently and are not stages of this pipeline.

Table C.2: Sampled problem families, transcribed from the generator sources named in the last column (code constants, not run outputs).

PDE	equation and boundary	sampled family (transcribed from the generator source)	source
Burgers 2D	$u_t + u(u_x + u_y) = \nu \Delta u$ on $(0, 1)^2$, $u = 0$ on $\partial\Omega$	$u_0 = a \exp(- x - c ^2/2w^2)$, $c_i \sim U(0.15, 0.85)$, $w \sim U(0.05, 0.20)$, $a \sim U(0.5, 2)$; $\nu \sim \log U(0.01, 0.1)$; outputs $t \in \{0.05, \dots, 0.25\}$	experiments/ mr-burgers2d/ engines.py: params_draw
Poisson 2D	$-\Delta u = f$ on $(0, 1)^2$, $u = 0$ on $\partial\Omega$	$f = a \exp(- x - c ^2/2w^2)$, $c_i \sim U(0.15, 0.85)$, $w \sim \log U(0.02, 0.1)$, $a \sim U(0.5, 2)$	multistage- precision/ms_ parametric.py: sample_params
Poisson, L-shape	same source family on $(0, 1)^2 \setminus [\frac{1}{2}, 1]^2$	as above; sources centred in the removed quadrant rejected	experiments/ lshape
Heat 2D	$u_t = \kappa \Delta u$ on $(0, 1)^2$, $\kappa = 0.02$ fixed	polynomial-boundary Gaussian family (sin- gle_bc_poly_gaussian_v1); Crank–Nicolson $\Delta t = 0.025$	checkpoint_expanded_ seed790715 (lineage: linear-bank cell, job 3511417); printed CG measurements: job 3529772
Wave 2D (reflective)	$u_{tt} = c^2 \Delta u$ on $(0, 1)^2$, $u = 0$ on $\partial\Omega$	compact bump $\times$ Gaussian: half-widths $s_i \sim U(0.36, 0.42)$, centre $c_i \sim U(s_i + 0.025, 1 - s_i - 0.025)$, amplitude $\sim U(0.7, 1.3)$, $\sigma_i \sim U(0.12, 0.16)$, advective velocity $v_i \sim U(-0.5, 0.5)$ (zero every fourth case); speed $c \sim U(0.85, 1.15)$	experiments/ multiresolution- wave/audit_ dynamics.py: parameter_rows

C.1 HEADLINE SETTINGS, TIMES AND TIMING PROTOCOL

Table C.3 gives the settings and measured times behind every row of Table 1. Every cost and error come from the same invocation. Timings use GPU warm-up and synchronisation, double precision and the highest matrix-multiplication precision; all repetitions, including outliers, enter the median, and no time is borrowed from another allocation. Requested full fields and initialisation are charged; host transfer is included only where the table says complete query; for Burgers at 4096², copying the six double-precision output fields to the host adds about 246 ms to every arm. The Poisson and three-dimensional heat CG comparators are unpreconditioned; heat CG is warm-started from the previous time level. The Heat2D rows use checkpoint `expanded_seed790715`; Table C.1 separates that checkpoint’s lineage from the job that produced the printed measurements.

Table C.3: Supporting data for Table 1: the two settings of each frozen model, median times (ms), timing scope, allocation and evidence status. “EQ” and “dense” name the Burgers residual evaluation; “single” marks models measured at one setting. Where a run recorded both timing scopes, the speedups in the scope not used by Table 1 are listed; a series never mixes scopes. For heat that column gives the speedups against the lane’s tighter CN–CG setting (rtol 10^{−6}). * not in Table 1: a re-measurement of the row above it, or a development-source run at a mesh whose held-out row is in Table 1.

Problem	Mesh	Accurate	Fast	Accurate ms	Fast ms	FOM ms	Timing	Other scope or FOM: acc. / fast	Job	Status
Poisson 2D	256 ²	$q = 256$	$q = 0$	7.12	6.70	22.29	GPU query	—	3780692	development
Poisson 2D	1024 ²	$q = 256$	$q = 0$	4.32	3.94	60.02	GPU query	—	3783813	development
Poisson 2D	2048 ²	$q = 256$	$q = 0$	5.47	5.35	397.80	GPU query	28.7 × / 28.8 × (complete query)	4049279	development
Poisson 2D*	4096 ²	$q = 256$	$q = 0$	16.92	16.74	2472.81	GPU query	41.1 × / 41.7 × (complete query)	4051236	development
Poisson 2D*	4096 ²	$q = 256$	$q = 0$	17.17	17.01	2465.60	GPU query	37.6 × / 37.9 × (complete query)	4071227	development
Poisson, L-shape 2D	256 ²	$q = 64$	$q = 0$	3.03	2.88	11.64	complete query	—	3784663	development
Poisson, L-shape 2D	512 ²	$q = 64$	$q = 0$	4.80	4.72	29.13	complete query	—	3789568	development
Poisson, L-shape 2D	1024 ²	$q = 128$	$q = 0$	5.81	5.67	56.72	complete query	16.0 × / 16.5 × (GPU query)	4057694	development
Poisson, L-shape 2D	2048 ²	$q = 128$	$q = 0$	18.75	18.66	362.30	complete query	33.9 × / 34.3 × (GPU query)	4057694	development
Heat 2D	64 ²	—	single	—	11.23	2.50	GPU query	—	3529772	development
Heat 2D	256 ²	—	single	—	11.34	6.94	GPU query	—	3529772	development
Heat 2D	1024 ²	—	single	—	12.21	59.18	GPU query	—	3529772	development
Heat (wide bank) 2D	1024 ²	$q = 32$	$q = 0$	31.20	36.89	66.84	GPU query	6.33 × / 5.35 × (vs. CN–CG rtol 10 ^{−6})	4054669	development
Heat (wide bank) 2D	2048 ²	$q = 32$	$q = 0$	27.97	28.42	272.55	GPU query	28.8 × / 28.3 × (vs. CN–CG rtol 10 ^{−6})	4060928	sealed held-out
Heat (wide bank) 2D	4096 ²	$q = 32$	$q = 0$	31.68	32.19	1139.83	GPU query	149 × / 146 × (vs. CN–CG rtol 10 ^{−6})	4060928	sealed held-out
Heat (wide bank, batched fit) 2D	2048 ²	$q = 32$	$q = 0$	4.54	4.78	272.55	GPU query	177 × / 169 × (vs. CN–CG rtol 10 ^{−6})	4060928	sealed held-out
Heat (wide bank, batched fit) 2D	4096 ²	$q = 32$	$q = 0$	11.03	11.33	1139.83	GPU query	428 × / 416 × (vs. CN–CG rtol 10 ^{−6})	4060928	sealed held-out
Burgers 2D	256 ²	$q = 256$, EQ	$q = 0$, EQ	746.02	40.36	31.79	GPU query	—	3789570	development
Burgers 2D	512 ²	$q = 256$, EQ	$q = 0$, EQ	783.33	40.49	32.92	GPU query	—	3805065	development
Burgers 2D	1024 ²	$q = 256$, dense	$q = 0$, EQ	22053.85	40.27	81.31	GPU query	—	3789572	development
Burgers 2D	2048 ²	$q = 256$, EQ $M = 544$	$q = 0$, EQ	124.15	29.48	122.76	GPU query	0.99 × / 2.01 × (complete query); 6.07 × / 25.5 × (vs. tight Newton)	4071616	development
Burgers 2D	4096 ²	$q = 256$, EQ $M = 1088$	$q = 0$, EQ	107.51	40.73	523.85	GPU query	2.18 × / 2.68 × (complete query); 30.4 × / 80.3 × (vs. tight Newton)	4079320	development
Burgers (held-out cases) 2D	2048 ²	$q = 256$, EQ $M = 1088$	$q = 0$, EQ	114.91	29.19	164.43	GPU query	1.28 × / 2.49 × (complete query); 6.48 × / 25.5 × (vs. tight Newton)	4077566	held-out
Burgers (held-out cases) 2D	4096 ²	$q = 256$, EQ $M = 1088$	$q = 0$, EQ	99.96	40.60	537.39	GPU query	2.28 × / 2.74 × (complete query); 32.3 × / 79.6 × (vs. tight Newton)	4079321	held-out
Burgers (earlier model) 2D	1024 ²	—	single	—	41.68	68.04	GPU query	—	3534502	development
Burgers (earlier model) 2D	1024 ²	—	single	—	41.68	68.75	GPU query	—	3534502	development
Poisson 3D	32 ³	$q = 96$	$q = 0$	2.63	2.49	2.47	GPU query	—	4028642	accepted final
Poisson 3D	64 ³	$q = 96$	$q = 0$	3.36	3.22	4.46	GPU query	—	4028642	accepted final
Poisson (dev. sources) 3D*	128 ³	$q = 96$	$q = 0$	1.43	1.38	2.88	GPU query	1.57 × / 1.56 × (complete query)	4051032	development
Poisson (dev. sources) 3D	128 ³	$q = 96$	$q = 0$	1.86	1.80	12.55	GPU query	2.70 × / 2.66 × (complete query)	4051032	development
Poisson (dev. sources) 3D*	128 ³	$q = 96$	$q = 0$	1.90	1.76	12.40	GPU query	2.65 × / 2.70 × (complete query)	4056288	development
Poisson (dev. sources) 3D	256 ³	$q = 96$	$q = 0$	6.08	5.95	140.87	GPU query	4.18 × / 4.21 × (complete query)	4056288	development
Heat 3D	32 ³	$q = 96$	$q = 0$	77.78	38.35	5.16	GPU query	—	4033346	accepted final
Heat 3D	64 ³	$q = 96$	$q = 0$	85.44	40.38	10.84	GPU query	—	4033346	accepted final

Table C.4: Heat at high resolution, one frozen model per block (median times, ms). Errors are worst same-grid relative L^2 over all output times and over evolved times. Speedup uses the FOM rule of Table 1; the last column uses CN–CG at $\Delta t = 0.025$, rtol 10^{−6}. CN: reduced Crank–Nicolson steps; batched fit: each output time fitted independently to exactly propagated test moments (linear autonomous problems with eigenfunction tests only). The sealed 2D cohort was opened once; the Table 1 1024² row uses the development cases. Wide bank: $R = 128$, $k = 8$; earlier: the $R = 32$ checkpoint of the Heat rows; 3D: the Heat 3D model of Table 1. The 3D rows miss the 1 % all-times target; 256³ is the first 16 final cases, not a mesh trend.

Model	Mesh	Cohort	Stepping	q acc./fast	Err. all (%)	Err. evol. (%)	ms	FOM ms	CN–CG FOM	Speedup	vs. 10 ^{−6}
Wide bank	1024 ²	development (12)	CN	32 / 0	0.24 / 0.56	0.17 / 0.46	31.2 / 36.9	66.8	$\Delta t = 0.05$, rtol 10 ^{−3}	2.14 / 1.81 ×	6.33 / 5.35 ×
	1024 ²	sealed (16)	CN	32 / 0	0.49 / 1.36	0.22 / 0.84	27.3 / 27.8	43.5	$\Delta t = 0.05$, rtol 10 ^{−3}	1.59 / 1.57 ×	4.70 / 4.62 ×
	1024 ²	sealed (16)	batched fit	32 / 0	0.49 / 1.36	0.21 / 0.84	3.1 / 3.4	43.5	$\Delta t = 0.05$, rtol 10 ^{−3}	13.9 / 13.0 ×	41.0 / 38.3 ×
	2048 ²	sealed (16)	CN	32 / 0	0.49 / 1.36	0.22 / 0.84	28.0 / 28.4	272.5	$\Delta t = 0.05$, rtol 10 ^{−3}	9.74 / 9.59 ×	28.8 / 28.3 ×
	2048 ²	sealed (16)	batched fit	32 / 0	0.49 / 1.36	0.21 / 0.84	4.5 / 4.8	272.5	$\Delta t = 0.05$, rtol 10 ^{−3}	60.0 / 57.0 ×	177 / 169 ×
	4096 ²	sealed (16)	CN	32 / 0	0.49 / 1.36	0.22 / 0.84	31.7 / 32.2	1139.8	$\Delta t = 0.05$, rtol 10 ^{−2}	36.0 / 35.4 ×	149 / 146 ×
Earlier ($R = 32$)	4096 ²	sealed (16)	batched fit	32 / 0	0.49 / 1.36	0.21 / 0.84	11.0 / 11.3	1139.8	$\Delta t = 0.05$, rtol 10 ^{−2}	103 / 101 ×	428 / 416 ×
	2048 ²	development (12)	CN	24 / 0	1.68 / 4.56	1.11 / 4.56	15.9 / 8.9	173.1	$\Delta t = 0.1$, rtol 10 ^{−2}	10.9 / 19.5 ×	51.1 / 91.4 ×
	4096 ²	development (12)	CN	24 / 0	1.68 / 4.56	1.11 / 4.56	18.0 / 10.8	1014.6	$\Delta t = 0.1$, rtol 10 ^{−2}	56.5 / 93.7 ×	267 / 442 ×
3D model	128 ³	final (64)	CN	96 / 0	1.93 / 3.18	0.75 / 1.54	35.1 / 16.5	7.0	$\Delta t = 0.1$, rtol 10 ^{−2}	0.20 / 0.43 ×	0.83 / 1.78 ×
	128 ³	final (64)	batched fit	96 / 0	1.93 / 3.18	0.72 / 1.36	5.2 / 4.0	7.0	$\Delta t = 0.1$, rtol 10 ^{−2}	1.36 / 1.77 ×	5.67 / 7.40 ×
	256 ³	final, 1st 16 (16)	CN	96 / 0	1.27 / 2.02	0.55 / 0.91	37.0 / 21.0	173.3	$\Delta t = 0.05$, rtol 10 ^{−4}	4.69 / 8.27 ×	8.96 / 15.8 ×
	256 ³	final, 1st 16 (16)	batched fit	96 / 0	1.27 / 2.02	0.50 / 0.86	9.6 / 8.4	173.3	$\Delta t = 0.05$, rtol 10 ^{−4}	18.1 / 20.5 ×	34.7 / 39.3 ×

Three-dimensional Poisson ($-\Delta u = f$, Gaussian sources), heat ($u_t = \nu \Delta u$, Gaussian initial fields) and scalar Burgers ($u_t + u(u_x + u_y + u_z) = \nu \Delta u$, backward Euler, sign-upwind) are posed on the unit cube with zero Dirichlet walls and extend the operators of Appendix A across three axes. Navier–Stokes uses periodic vector fields, an incompressibility projection and the CNAB2 integrator; its FOM is not a CG solve. All four three-dimensional problems are evaluated on held-out final cohorts whose settings were frozen beforehand.

Table C.5: Three-dimensional configurations, read from the run records. Domains are the unit cube with homogeneous Dirichlet data, except Navier–Stokes, which is periodic. Burgers and Navier–Stokes appear in Table 3.

Problem	Mesh	Cases	Correction ranks	Named FOM	Residual	Allocation
Burgers 3D	33^3 nodes	32 final	0, 192	Newton–BiCGStab, $\Delta t = 0.01$	dense	job 4021709
Poisson 3D	$32^3, 64^3$	64 final	0, 32, 96 ($k = 16$)	CG, $\text{rtol } 10^{-2}$, no preconditioner	dense	job 4028642
Heat 3D	$32^3, 64^3$	64 final	0, 32, 64, 96 ($k = 32$)	CN–CG, $\Delta t = 0.05$, $\text{rtol } 10^{-4}$, warm start	dense	job 4033346
Navier–Stokes 3D	32^3 periodic	32 final	0, 32, 64, 128, 256 ($k = 64$, $R = 1536$, $M = 2048$)	CNAB2, $\Delta t = 0.01$	dense	job 4027788

C.2 SOURCE RECORDS

The accompanying source package retains the full experiment archive, including failed configurations, retractions, training records, checkpoints, case-level fields, timing repetitions and independent audits. The PDF contains the evidence needed for its stated claims rather than every experiment in that archive. Table generators and frozen evidence identify the displayed values through these machine-readable manifests:

- `tables/provenance.json`: original two-dimensional studies.
- `tables/rewrite-provenance.json`: paired CG comparisons.
- `tables/burgers-iterative-provenance.json`: the earlier fine-grid Burgers tight and relaxed controls.
- `tables/headline-provenance.json`: every row of Tables 1, C.3, 3 and C.5 and every point of Figure 1, with the commit and SHA256 of each source record.

Each manifest records source hashes; its corresponding evidence retains solver configuration, checkpoint identity and allocation metadata. Development selection and final evaluation are distinct. Solver exit flags document numerical stopping, not a proof of global optimality.

D VALIDATION OF THE CORRECTION AND QUADRATURE STUDIES

Table D.1: The sealed cohort (lane b-seeds, job 3804465, NVIDIA A100-PCIE-40GB), opened once after every choice was frozen. Top: per rung of the dense $M = 4(K + q)$ ladder, the incumbent’s sealed worst evolved error and device time, the seed mean $\pm$ sample standard deviation on the sealed and the development cohort, the lane’s sealed-over-development ratio of seed means (raw and difficulty-normalised) and the converged count over the 4 checkpoints. Bottom: per-checkpoint verdicts. The incumbent’s $q = 0$ rung is a single sealed case converged to a wrong branch (gradient exit, zero budget exits); the pre-registered ratio criterion fails at $q = 0$ for the incumbent alone (5.35) and the convergence criterion fails on `seed2` at $q = 64$ (3 budget exits).

q	M	incumbent sealed %	ms	seeds sealed %	seeds development %	sealed / dev	normalised	converged
0	64	10.1120	375.1	2.8023 ± 0.6357	2.0002 ± 0.4986	1.40	1.41	4 of 4
16	128	1.4617	480.8	1.7305 ± 0.1297	1.5988 ± 0.0995	1.08	1.09	4 of 4
32	192	1.4458	540.5	1.5044 ± 0.1422	1.3727 ± 0.1314	1.10	1.14	4 of 4
64	320	1.3252	754.5	1.3528 ± 0.1099	1.2130 ± 0.0580	1.12	1.13	3 of 4
128	576	1.0964	1460.4	1.0461 ± 0.0619	0.9562 ± 0.0264	1.09	1.17	4 of 4
256	1088	0.6789	6725.5	0.6188 ± 0.0345	0.4987 ± 0.1164	1.24	1.20	4 of 4

checkpoint	job	monotone (evolved)	monotone (all-times)	every rung converged	error span	cost span	knob bar
incumbent	3804465	yes	yes	yes	$14.89 \times$	$17.93 \times$	yes
seed1	3804465	yes	yes	yes	$3.52 \times$	$13.25 \times$	yes
seed2	3804465	yes	yes	no	$4.81 \times$	$13.17 \times$	no
seed3	3804465	yes	yes	yes	$5.22 \times$	$14.62 \times$	yes

Reading the validation tables. Correction rank q is the number of added coefficient directions; M is the number of weak test modes and m the quadrature node count. A scheduled ladder changes M with q ; the main fixed-test-space study holds M constant. A sealed cohort is opened only after choices are frozen. Confirmed quadrature passes every independent reconstruction of the rule; marginal means some reconstructions fail. These statuses concern rule construction, not just the error of one selected trajectory. Same-grid error and error against a refined reference measure different quantities.

Table D.2: The EQ ladder with the cheapest rule passing the primary bar in its draw per rung, timed in one allocation (job 3780164, A100 80 GB), with its same-job dense twins, and the construction status from the four-draw replication (job 3783811): confirmed = every re-draw passes, marginal = only the listed fraction does. Errors are as measured with these rules (SHA256-identified); they are not properties of the construction above $q = 32$. Ladder rows read from the lane summary.json (final).

q	m	fit states	$\rho_{\max}$ (this draw)	construction status	EQ evolved %	EQ all %	EQ ms	dense evolved %	dense ms	EQ/dense cost
0	1024	64	0.0153	confirmed (3 of 3 re-draws)	1.8891	2.5629	59.1	1.8890	292.9	0.202
16	1024	64	0.0935	confirmed (3 of 3 re-draws)	1.4270	2.4806	80.6	—	—	—
32	1024	42	0.0533	confirmed (2 of 2 re-draws)	1.2493	2.3534	97.7	—	—	—
64	1024	25	0.0531	marginal (2 of 6 draws pass)	1.2275	2.1489	120.6	1.0843	624.9	0.193
128	2048	64	0.0669	marginal (4 of 5 draws pass)	0.8936	1.8116	246.9	0.8930	1190.5	0.207
256	2048	64	0.1074	marginal (1 of 5 draws pass)	0.5389	0.9053	722.2	0.5194	3915.1	0.184

Table D.3: How the corrections are solved on Burgers, one job (job 3734098, NVIDIA A100 80GB PCIe): joint LM on (z, y) , block-damped, and plain variable projection at $q = 64$ and 128. Same error where all converge; the block-damped step is the one kept.

arm	M	worst all-times %	budget exits	all stationary	device ms
$q=64$, joint LM on (z, y)	320	2.1489	6	no	1119.7
$q=64$, block-damped	320	2.1489	0	yes	619.2
$q=64$, plain variable projection	320	2.1489	0	yes	9527.9
$q=128$, plain variable projection	256	1.8116	297	no	47566.9
$q=128$, block-damped	256	1.8116	0	yes	825.5

Table D.4: Solver-side controls on Burgers2D, 32 held-out validation cases, one allocation: stopping tolerance, EQ node count m and iteration cap, with full-order settings for scale.

setting	worst % (32 held-out)	device ms	early-stopped
EQ $m=256$, tol 10^{-6} , cap 180	7.245	51.0	0/96
EQ $m=512$, tol 10^{-8}	6.712	64.7	0/96
EQ $m=512$, tol 10^{-3}	6.701	41.6	0/96
EQ $m=512$, cap 2 (early-stopped)	90.324	28.4	96/96
FOM Newton 10^{-2} , $\Delta t=0.01$	6.807	9.9	0/96
FOM Newton 10^{-2} , $\Delta t=0.005$	35.357	17.4	0/96
FOM Newton 10^{-4} , $\Delta t=0.005$	6.171	22.4	0/96
FOM Newton 10^{-6} , $\Delta t=0.005$	6.172	88.3	0/96
FOM 128^2 , $\Delta t=0.005$	9.849	21.2	0/96
FOM 64^2 , $\Delta t=0.01$	16.095	14.0	0/96